

Definition of Carbs:

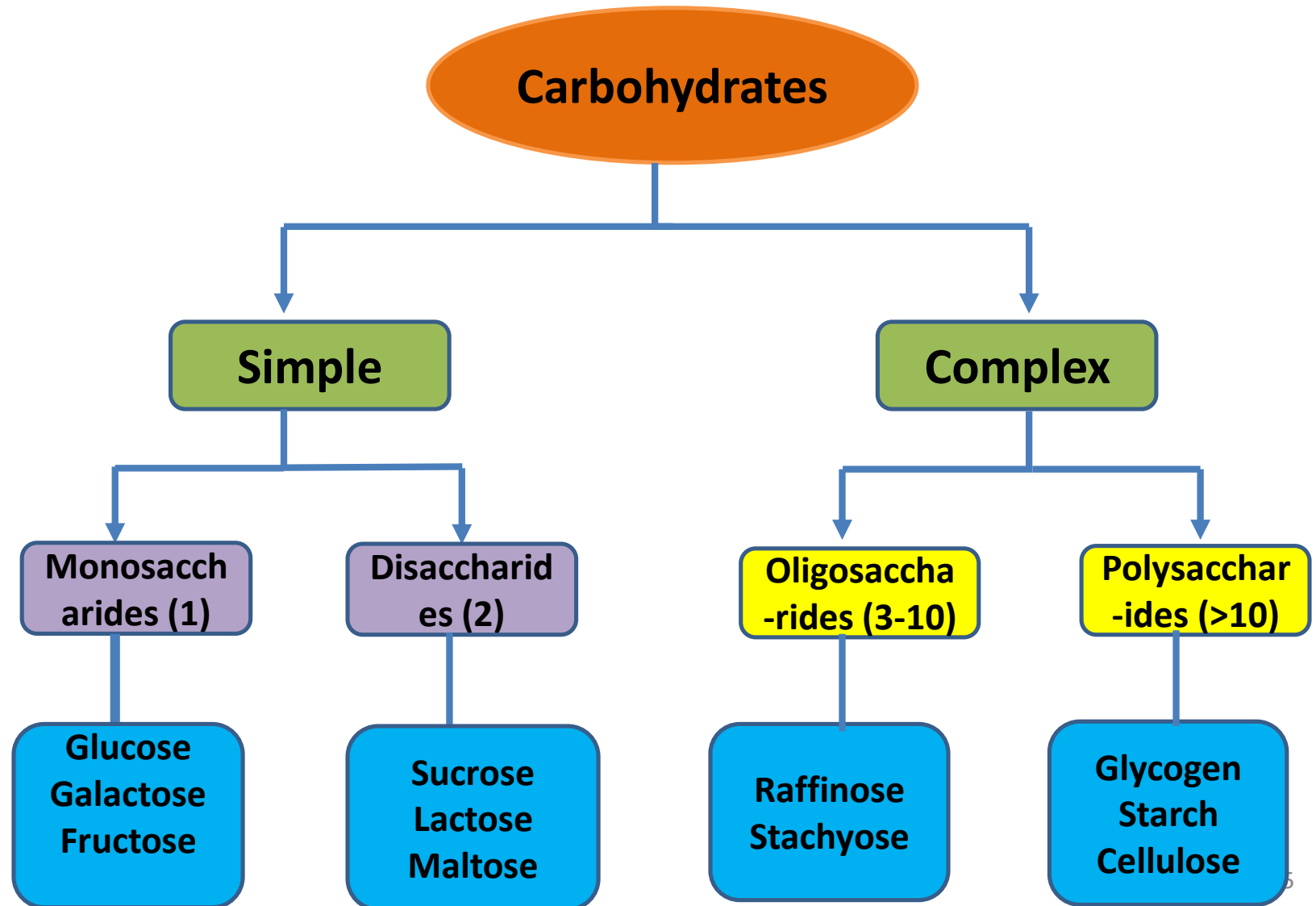
- Carbohydrates are the most abundant organic molecules found in nature.
- Structurally defined as **aldehydes** or **ketones** with **multiple hydroxyl** groups, or “hydrate” of carbon (– C-H₂O).
- The term **Carbohydrate** is the most common in biochemistry, where it is a synonym of **saccharide**, which means **sugar**.

Main Functions of Carbohydrates

1. **Energy Production**, the primary role of carbohydrates is to supply energy to all cells in the body.
2. **Energy Storage**, the excess glucose is stored as glycogen (the majority of which is stored in the muscles and liver).
3. **Building Macromolecules**, ribose and deoxyribose, which are essential building blocks of important macromolecules, such as RNA, DNA, and ATP.

- 4. Sparing Protein**, the presence of adequate glucose basically spares the breakdown of proteins (primarily of muscle tissue) from being used to make glucose needed by the body.
- 5. Lipid Metabolism**, as blood-glucose levels rise, the use of lipids as an energy source is inhibited, and adequate glucose levels in the blood also prevent the development of ketosis.
- 6. Antigenic** – ABO blood groups
- 7. Structural components** of cell walls in plants and insects, cartilage in vertebrates.

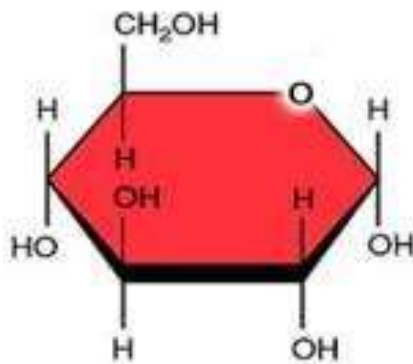
Classification of Carbohydrates



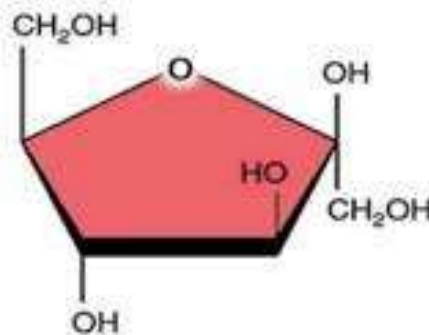
1- Monosaccharides:

- Are the simplest carbohydrates in that they **cannot be hydrolyzed** to smaller carbohydrates.
- Monosaccharides consist of:
 - **3-6 carbon** atoms.
 - Trioses (dihydroxy acetone)
 - Tetroses (erythrose)
 - Pentoses (ribose)
 - Hexoses (glucose)
 - And a carbonyl group aldehyde(aldose) or ketone(ketose), with several hydroxyl groups.

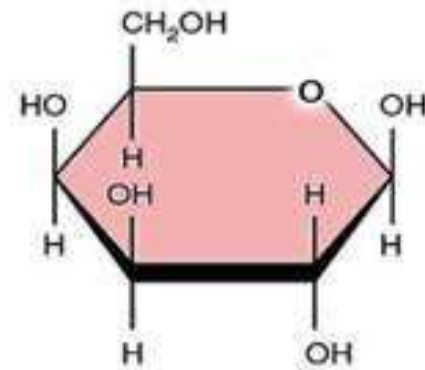
➤ Examples of monosaccharides:



Glucose



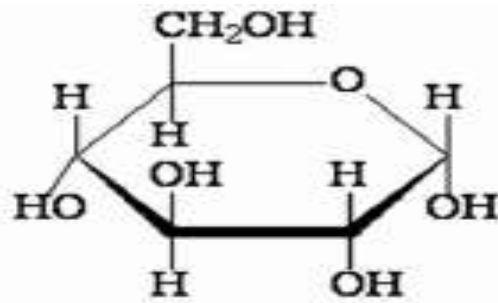
Fructose



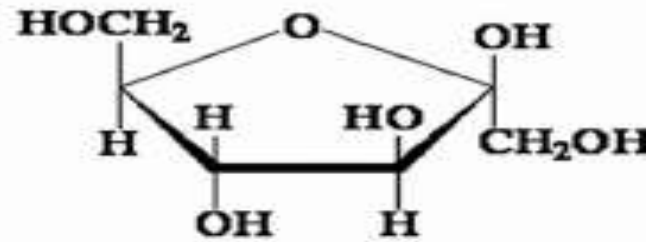
Galactose

Aldoses & Ketoses

A

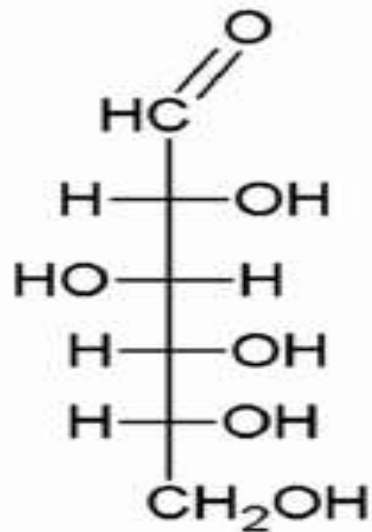


Glucose

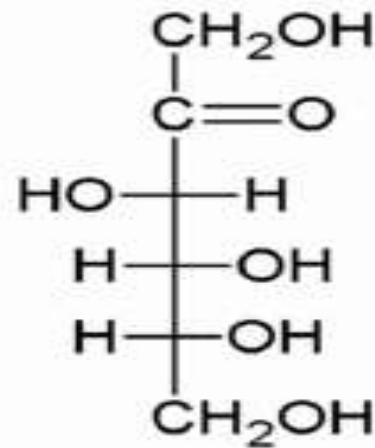


Fructose

B



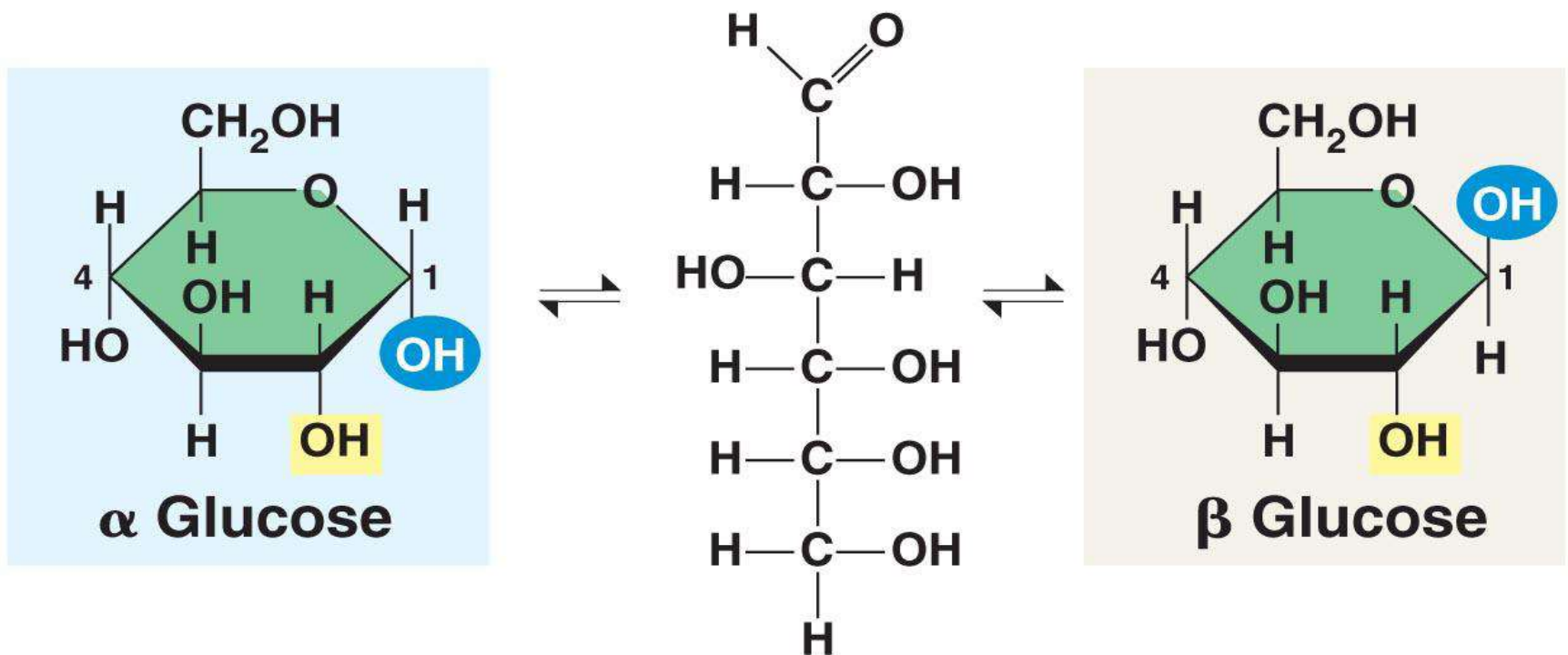
Glucose



Fructose

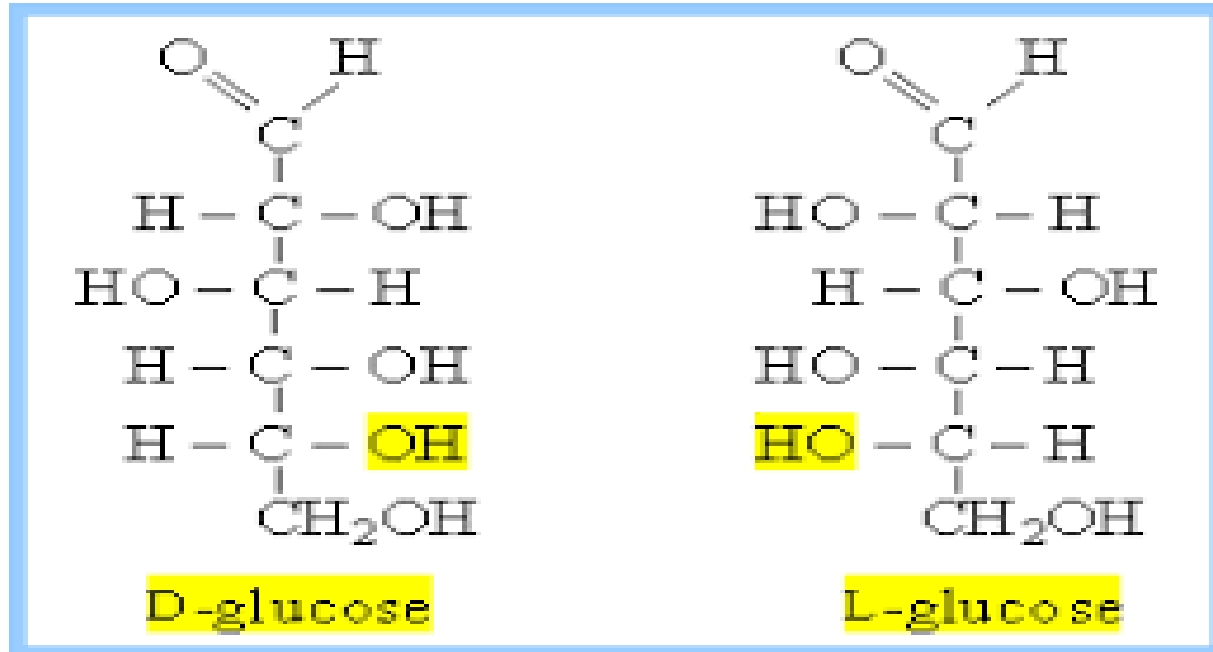
Differences between α and β sugars

The **cyclic forms** of carbohydrates can exist in α or β forms, based on the **position** of the ($-\text{OH}$) or ($-\text{CH}_2\text{OH}$) group at the **anomeric carbon**.



Differences between D and L sugars

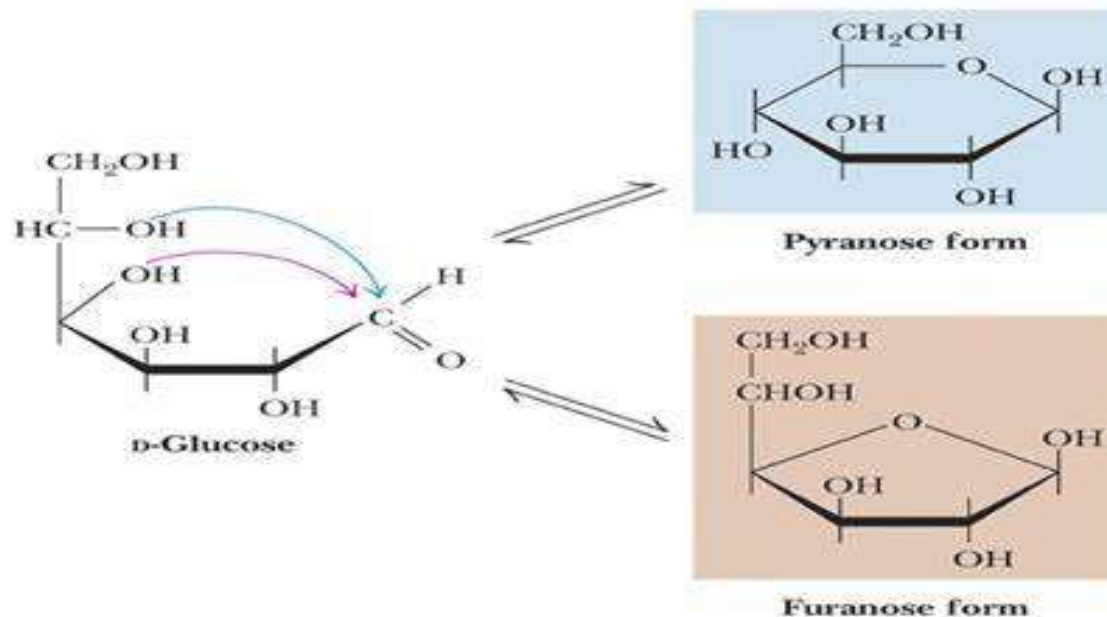
These two forms are **enantiomers**, their molecular structures are mirror images of each other. In **D-(Dextro)** form the **(-OH)** of the **penultimate carbon atom** on the **right side**, while in **L-(Levo)** form the **(-OH)** in the **left side**.



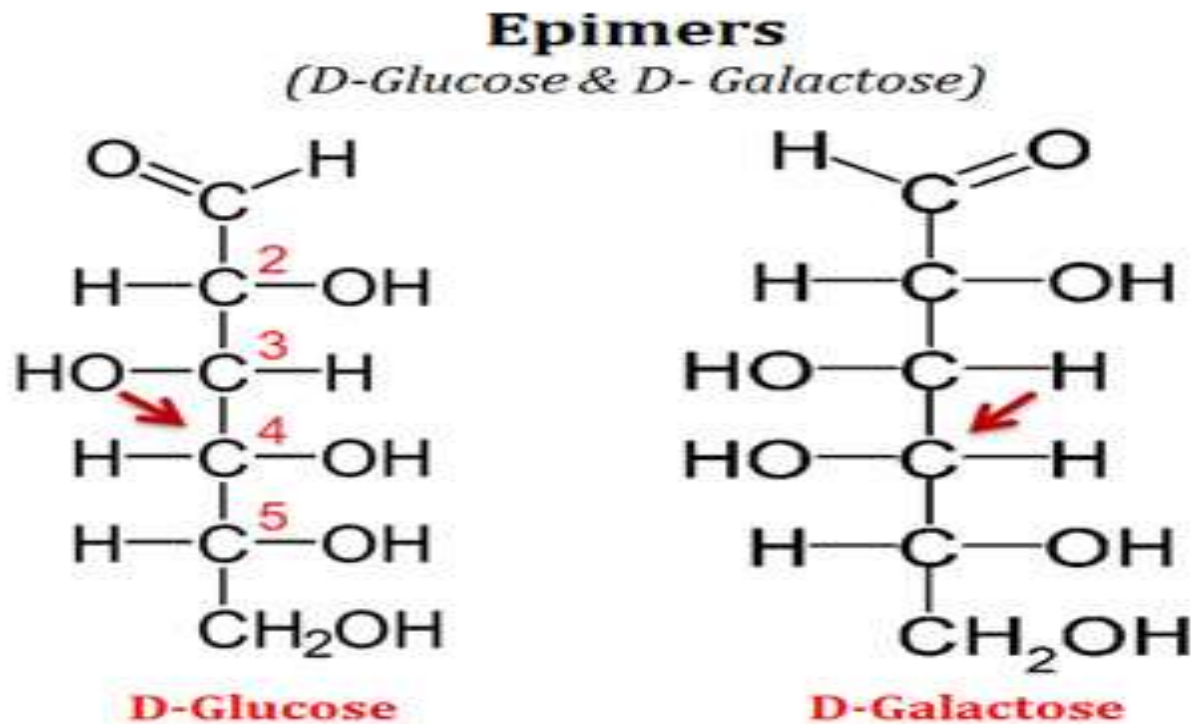
What are Pyranose and Furanose?

Pyranose is a term for saccharides that have a six-membered ring structure consisting of five carbon atoms and one oxygen atom, while

Furanose have a five-membered ring structure consisting of four carbon atoms and one oxygen atom.



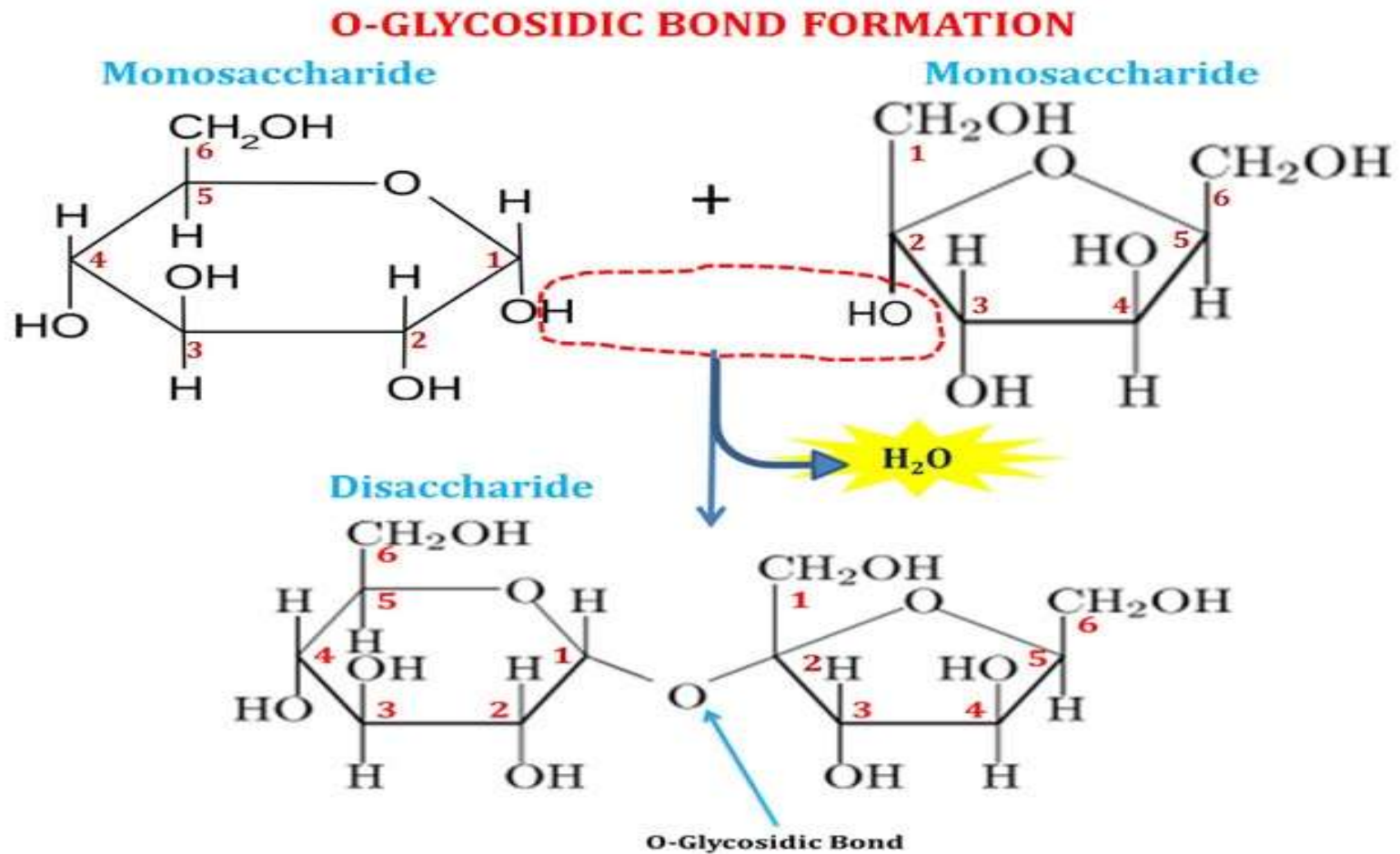
What are Epimers? Isomers that differ in configuration around only one specific carbon atom are defined as **epimers** of each other.



2- Disaccharides:

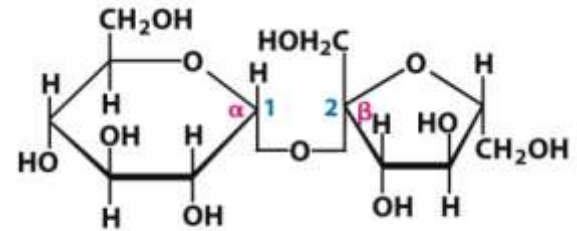
- Are formed as a result of combination of 2 monosaccharide units by glycosidic bond.
- A glycosidic bond is formed by a condensation reaction, releasing water molecule.
- They can yield two monosaccharides on hydrolysis.

➤ Mechanism of glycosidic bond formation:



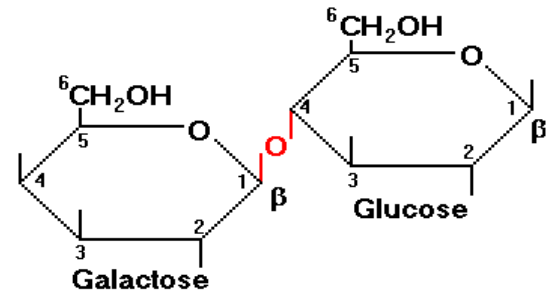
➤ Examples of disaccharides:

Sucrose (“table sugar”)
glucose + fructose

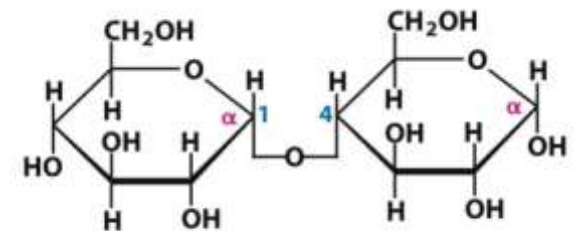


Sucrose
(α -D-Glucopyranosyl-(1 $\rightarrow$ 2)- β -D-fructofuranose)

Lactose (“milk sugar”)
galactose + glucose



Maltose (“malt sugar”)
glucose + glucose



Maltose
(α -D-Glucopyranosyl-(1 $\rightarrow$ 4)- α -D-glucopyranose)

3- Oligosaccharides:

- Short carbohydrate chains of 3 – 10 monosaccharides.
- Found in legumes and human milk.
- Their main functions: strengthening immune system, reducing appetite, and improving gut health(as prebiotic).
- Examples of oligosaccharides:

○ Raffinose

○ Stachyose

cannot be broken down by human enzymes,
though can be digested by colonic bacteria.



Stachyose

4- Polysaccharides:

- Polysaccharides contain greater than 10 mono saccharide units.
- They represent an important class of biological polymers.
- Examples of polysaccharides:
 - Starch
 - Glycogen
 - Cellulose

Differences between polysaccharides

Polysaccharides	Sources	Structures	Digestions	Functions
Starch (amylopectin, & amylose)	Plants	Polymers of glucose(α 1-4, and α 1-6)	Digested by human enzymes	Major storage form in plants, and has nutritional function.
Glycogen	Animals	Polymers of glucose(α 1-4, and α 1-6), highly branched chain	Digested by human enzymes	Major storage form in animals
Cellulose	Plants	Polymers of glucose(β 1-4)	Can't be digested by human enzymes	Structural function

Polysaccharides, cont.

- Glycosaminoglycans, also known as mucopolysaccharides, made of repeating units of disaccharides containing a derivative of an amino sugar.
- One of the sugars in unit has negatively charged carboxylate or sulfate group.
- Usually attached to proteins.
 - proteoglycans

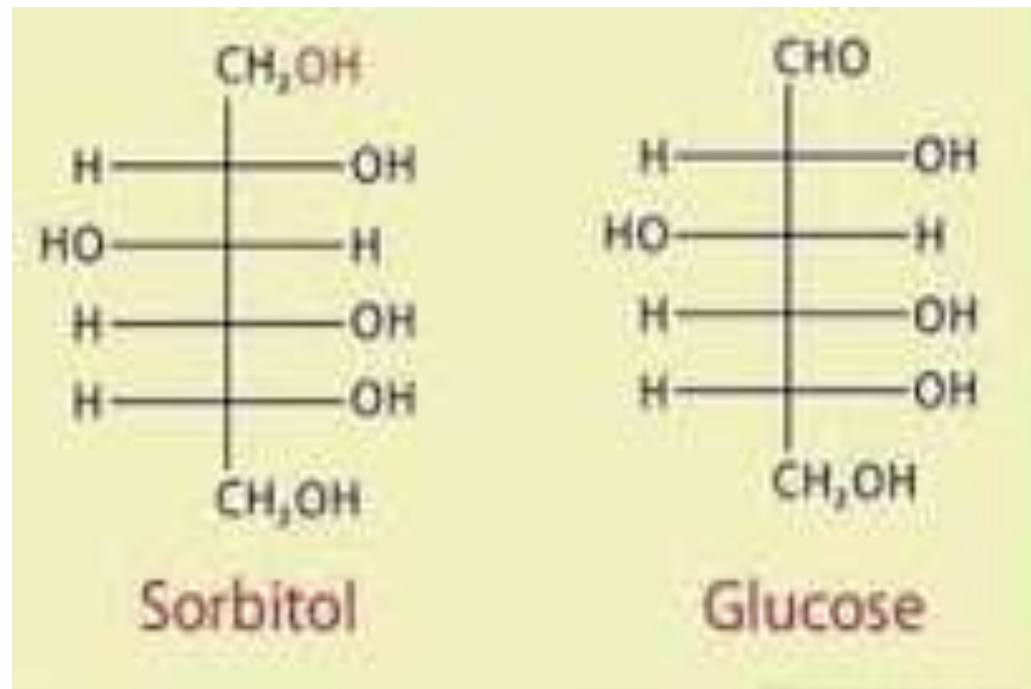
What are sugar alcohols?

- Sugar alcohols or polyols, are a type of carbs widely used as **low calorie** sweeteners.
- Are added to foods to give them a **sweet taste** as an alternative to regular sugar.
- Although they closely resemble the taste of sugar, but they **don't have the same negative effects** as regular sugar, such as:
 - promoting **tooth decay**.
 - significantly **raising blood sugar** levels.

- They are **low digestible** carbs,
- after eating this type of sugar the **small intestine doesn't completely absorb** them, instead,
- they travel to the large intestine, where **bacteria ferment** them.
- **Fiber** and **resistant starch** are also considered **low digestive carbs**.
- Similarly to fiber, certain sugar alcohols may contribute to a healthy digestive system by promoting the **growth of beneficial bacteria**.*

➤ Example of common sugar alcohols are:

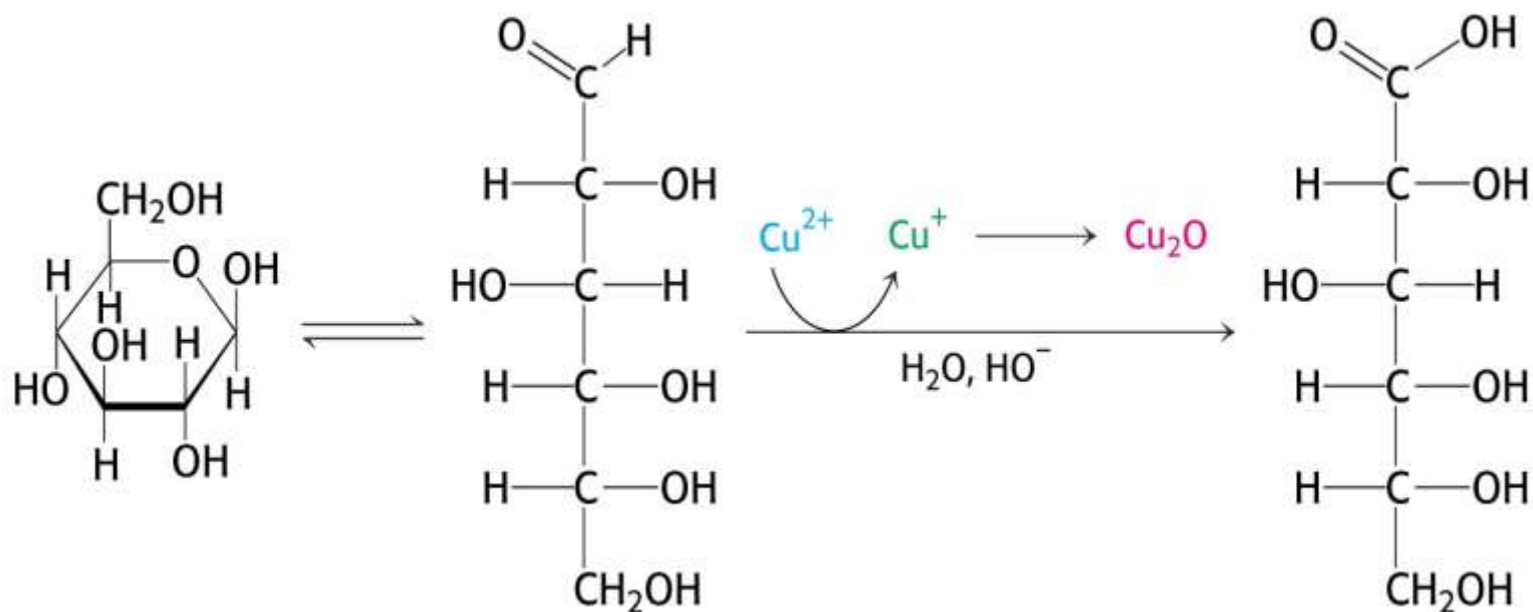
- mannitol
- maltitol
- sorbitol
- xylitol



Reducing Sugars & Non Reducing Sugars:

- A **reducing sugar** is any sugar that is capable of acting as a **reducing agent**.
- It has a **free aldehyde** group or a **free ketone** group.
- **All monosaccharides** are reducing sugars, along with **some** disaccharides, oligosaccharides, and polysaccharides.

- Examples of a reducing sugar that react with an oxidizing agent:

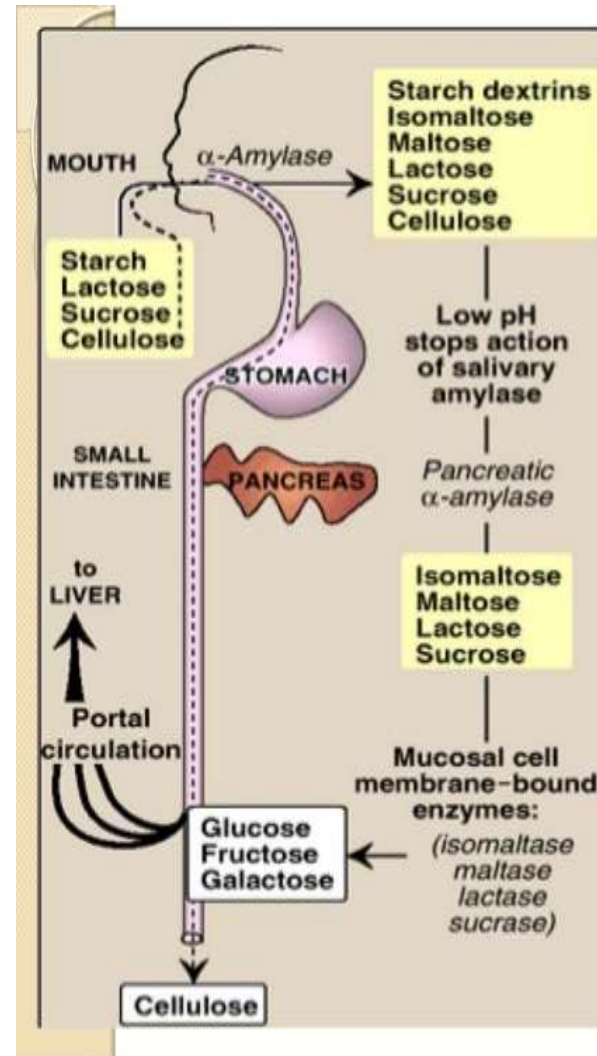


Glucose

Gluconic Acid

Digestion of Carbohydrates

- Digestion of carbohydrates begins in the **mouth**.
- The major dietary polysaccharide is of **plant** origin(starch).



Digestion of carbohydrate

- Salivary α - amylase hydrolyze α 1-4 bonds of dietary polysaccharides, resulting dextrin.
- Carbohydrate digestion halts temporarily in the stomach, because the high acidity inactivates the salivary α - amylase.
- In the small intestine, the acidic stomach neutralized by bicarbonate secreted by the pancreas, and pancreatic α - amylase continues the process of starch digestion.

➤ The final digestive processes occur at the **mucosal lining of the upper jejunum**, which includes the action of several disaccharidases.

○ isomaltose isomaltase → 2 glucose

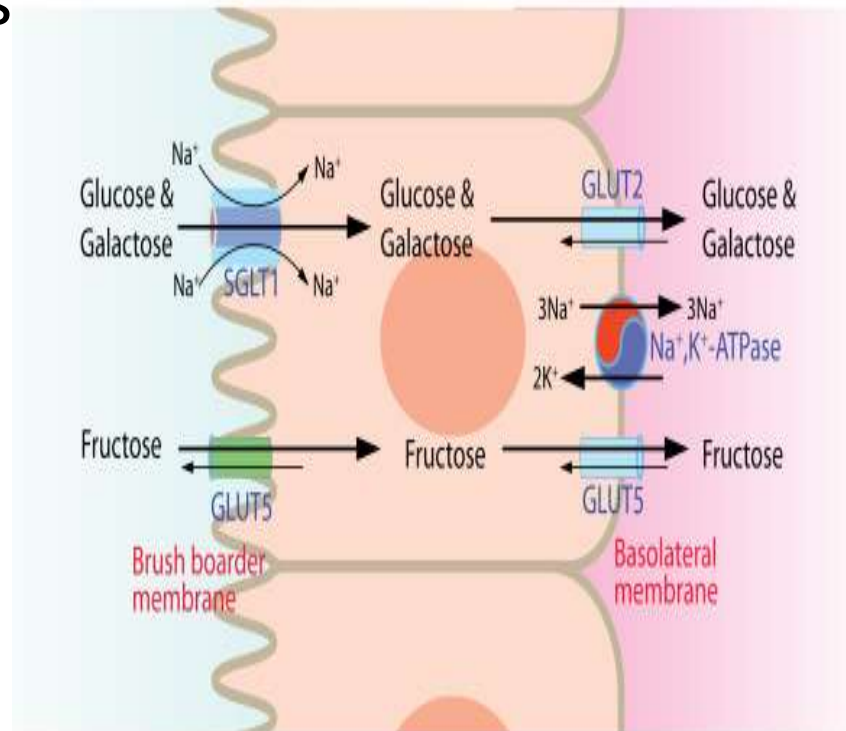
○ maltose maltase → 2 glucose

○ sucrose sucrase → glucose + fructose

○ lactose lactase → galactose + glucose

Absorption of Carbohydrates

- Most of the monosaccharides are absorbed from the **mucosal surface of jejunum** and **upper ileum**.
- **Glucose*** and **Galactose*** are absorbed into the epithelial cells lining the mucous membrane of small intestine by an **active transport** system sodium dependent glucose transporter-1 (**SGLT-1**).



- Fructose* is absorbed by facilitated diffusion (GLUT-5).
- Glucose and Galactose are transported from the epithelial cell into the interstitial space and hence to blood capillaries of portal system through facilitated diffusion by glucose transporter-2 (GLUT-2).

Abnormalities of Carbohydrate Digestion and Absorption

1. **Lactose intolerance:** refers to a condition in which lactose cannot be digested due to deficiency of enzyme lactase.
 - **Congenital lactase deficiency** occur in newborns.
 - **Secondary lactase deficiency** occur in adults, due to a variety of gastrointestinal diseases that damage the mucosa of the small intestine.

Symptoms:

Diarrhea, gases production, and bloating.

2. Disaccharidase deficiency syndrome:

- **Deficiency of disaccharidases (sucrase, lactase, & maltase)**, this decrease hydrolysis of disaccharides, hence decreases their absorption*, leading to increasing their concentration* in small intestine and large intestine.

Symptoms:

abdominal cramps, bloating, diarrhea, acidic stools, & explosive diarrhea.

3. Glucose-Galactose Malabsorption:

- A very rare **mutation of SGLT-1** results in a defective transporter that presents with severe diarrhea at birth.
- Small intestine ability to resorb glucose and galactose is decreased.

Symptoms:

similar to disaccharidase deficiency syndrome.

Glucose

- Glucose is a major circulatory carbohydrate and **energy source** for human body.
- It is derived primarily from **dietary carbohydrates***, **body stored glycogen*** (glycogenolysis), and from **non carbohydrate substances*** like **a. acids, glycerol**, and **lactic acid** (gluconeogenesis) which occur in **liver**.

Glycemic Index

- Glycemic Index (GI) is a relative ranking of carbohydrate in foods according to how they affect blood glucose levels.
- Carbohydrates with a low GI value (55 or less) are more slowly digested*, absorbed* and metabolized*,
- and cause a lower and slower rise in blood glucose and, therefore insulin levels.

Glycemic Index

GLYCEMIC Index	Rating	Food Item	Rating	Food Item	Rating	Food Item
	110	Maltose	62	Banana	36	Chick peas
	100	Glucose	59	Sucrose	36	Lima beans
	98	Potato, russet, baked	59	Sweet corn	36	Yogurt
	97	Parsnips	51	Bran	34	Pear
	92	Carrots	51	Green peas	34	Milk, whole
	87	Honey	51	Potato chips	32	Milk, skim
	80	Potato, instant mashed	51	Sweet potato	29	Kidney beans
	80	Cornflakes	50	White spaghetti	29	Lentils
	72	Whole-wheat bread	49	Oatmeal	26	Peach
	72	White rice	45	Grapes	26	Grapefruit
	69	White bread	42	Whole-wheat spaghetti	25	Plum
	67	Shredded wheat	40	Orange	23	Cherries
	66	Brown rice	39	Apple	20	Fructose
	64	Raisins	38	Tomato	15	Soybeans
	64	Beets	36	Ice cream	13	Peanuts

How Blood Glucose Regulated?

- The **liver** is the primary organ involved in the regulation of blood glucose, followed by peripheral tissues (**muscle and adipocytes**).
- The liver regulates blood glucose by:
 - extracting glucose from blood.
 - performing glycogenesis, glycogenolysis, & gluconeogenesis.
- While muscle & adipose tissue, they use glucose for their energy needs.

➤ The liver's **uptake** and **output** of glucose and the **use of glucose** by peripheral tissues depend on the physiological balance of several **hormones** that:

1- Lower blood glucose level (**insulin**).

2- Rise blood glucose level (**glucagon, epinephrine, GH, glucocorticoids,...**).

➤ Disorder of carbohydrate metabolism leads to a common endocrine disorder known as Diabetes Mellitus (DM),

- DM is a chronic disease characterized by **hyperglycemia** due to an absolute* or relative* **lack of insulin**, or **insulin resistance**.
- Is the leading cause of **adult blindness**, **renal failure**, **heart attacks**, and **strokes**.

DM is generally classified into **two** types:

- 1- Type I (Insulin dependent DM).
- 2- Type II (Non insulin dependent DM).

Differences between Type I & Type II DM

Type I DM

- 1- **Age** < 30 years
- 2- **Cause:** auto immune destruction of pancreatic islet cell.
- 3- **Insulin:** low or absent
- 4- **Weight:** normal or underweight.
- 5- Treatment: Insulin is necessary.
- 6- Epidemiology: 10-20%

Type II DM

- > 40 years
- Insulin Resistance
- normal or high
- overweight or obese
- changing life style, medications.
- 80-90%

Symptoms of DM:

- Excessive urination (polyuria).
- Excessive thirst (polydipsia).
- feeling very hungry (polyphagia).
- feeling very fatigued.
- blurry vision.
- cuts or sores that don't heal properly.

Mechanism of Main Symptoms of DM

1. Hyperglycemia: insulin is absent, deficient, or ineffective.

- decreased glucose uptake by muscle & adipose tissue.
- increased hepatic glucose production.
- reduced glycogenesis
- increased lipolysis & proteolysis

2- Polyuria (urine output > 3liters/day in adults):

- Hyperglycemia exceeds the renal threshold → glucose appears in urine (glycosuria).
- Glucose acts as an osmotic diuretic → pulls water with it into urine.

3- Polydipsia (excessive thirst):

- Hyperglycemia → osmotic diuresis → dehydration →
↑ plasma osmolality → stimulation of hypothalamic
thirst center → polydipsia

4- Polyphagia (excessive hunger):

- decreased glucose uptake by cells → cellular starvation → activation of hypothalamic feeding center

Important biochemical changes that occur in uncontrolled DM:

1. Hyperglycemia
2. Increase gluconeogenesis & lipolysis.
3. Ketone body formation.
4. Metabolic acidosis.
5. Proteolysis with negative nitrogen balance.
6. Dehydration
7. Electrolyte imbalance
8. Increased plasma osmolality

THANK YOU!

